Homogeneous and Heterogeneous Databases

1 Homogeneous Distributed Databases

In a **homogeneous distributed database system**, all sites have identical database management system (DBMS) software, are aware of one another, and agree to cooperate in processing users' requests.

Key Characteristics

- **Identical Software:** Every site uses the same DBMS software.
- **Global Awareness:** Sites are aware of each other and cooperate.
- **Reduced Autonomy:** Local sites surrender a portion of their autonomy (e.g., right to change schemas or software) to maintain system consistency.
- **Cooperative Processing:** Software cooperates to exchange information about transactions, enabling processing across multiple sites.

2 Heterogeneous Distributed Databases

In contrast, in a **heterogeneous distributed database**, different sites may use different schemas and different database-management system software.

Key Challenges

- **Diverse Software/Schemas:** Sites may use different DBMS software and data models.
- **Limited Awareness:** Sites may not be aware of one another.
- **Limited Cooperation:** Only basic facilities for cooperation in transaction processing might be provided.
- **Query Problems:** Differences in schemas are a major hindrance to query processing.
- **Transaction Hardship:** Divergence in software makes accessing multiple sites for a single transaction difficult.

3 Banking Database Example

The following schema represents a typical banking database used in distributed environments.

Figure 19.1: Banking Database Schema

Relational Schema

branch (<u>branch_name</u>, branch_city, assets)

account (<u>account_number</u>, branch_name, balance)

depositor (<u>customer_name, account_number</u>)

Most distributed database studies focus on homogeneous systems, though heterogeneous issues are critical for real-world integration.